

Redshift Reduction Telescope: Prototype presentation

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Why have the researchers built this?

The researchers have built this to aid in the reduction of cosmological redshift; it is described above, and is a serious problem in astronomical imaging. According to Jain, “enhance our *interpretation* of” redshift, removing it via software to allow for a nicer view of the universe. The researchers wanted to help astrophotographers and hobbyists who like taking pictures of the universe. Our prototype shall fix this using software reconstruction in the concise language Python, with the help of libraries such as OpenCV. This is a great tool for astrophotographers and hobbyists.

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$$v = H_0 \cdot d$$

$$v \approx (2.23 \cdot 10^{-18}) \cdot (2 \cdot 10^{24}) \text{m s}^{-1}$$

$$v \approx 4.46 \cdot 10^6 \text{m s}^{-1}$$

What is redshift? — continued

The velocity of the galaxy has been obtained. Now, using the relativistic redshift formula:

$$z = \left(1 + \frac{v}{c}\right) \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$
$$z \approx \left(1 + \frac{4.46 \cdot 10^6}{3 \cdot 10^8}\right) \frac{1}{\sqrt{1 - \frac{(4.46 \cdot 10^6)^2}{(3 \cdot 10^8)^2}}}$$
$$z \approx 1.015$$

Applying this to the original wavelength:

$$\lambda_{obs} = z \lambda_{emit}$$
$$\lambda_{obs} \approx (4 \cdot 10^{-7})(1.015)$$
$$\lambda_{obs} \approx 4.06 \cdot 10^{-7} \text{ m}$$

What is redshift? — continued²

Redshift, put simply, is but a distortion; a stretching of light, due to the expansion of the universe. The below diagram explains redshift concisely:

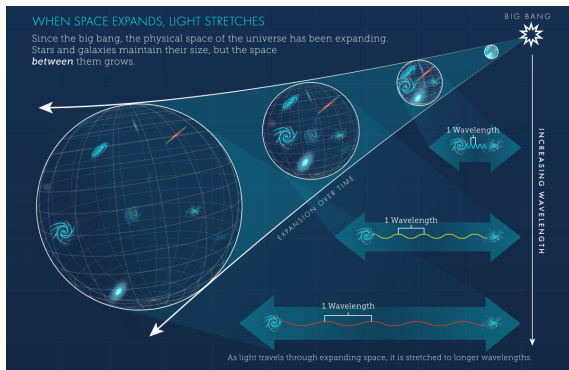


Figure: Redshift. Source: NASA, ESA, and Leah Hustak from STScI.

Mechanism and operation: building

The researchers shall use a standard refracting telescope; they shall buy a pre-built telescope, due to the relatively lower cost. T